

Memorandum for the Record

May 4, 2026

Subject: Bruker Titan 800 XRF Guns (Name Brand Justification)

In accordance with FAR 13.106-1(b)(1)(i), I have determined that there is only one responsible source (brand name), and no other suppliers or services will satisfy agency requirements.

Description of Action: Defense Logistics Agency (DLA) – Disposition Services proposes to enter a contract on a basis of other than full and open competition for the acquisition of three (3) Bruker Titan hand-held XRF spectrometers to support Strategic Materials Testing. The contract will be a firm-fixed-price contract under FAR 13.106-1(b)(1)(i).

Description of Supplies / Services: The purchase of three (3) Bruker Titan 800 XRF Guns. DLA Disposition Service currently owns two Titan XRF instruments manufactured by the Bruker Corporation. The two instruments were acquired by DLA Strategic Material and transferred to Disposition Services (Hand Receipt). DLA Disposition Services has a need for three additional Titan XRF Guns to place one at each DSD Region and one at the Disposition Services training center.

Reason for Authority Cited: In accordance with FAR 13.106-1(b)(1)(i), there is only one source (brand name) that can provide the supplies or services at the level of quality required because the supplies or services are unique or highly specialized.

DLA Disposition Services currently owns two Titan XRF instruments manufactured by Bruker. Since different manufacturers' instruments create instrument-based errors in data, Disposition Services would have inconsistent data if XRF instruments produced by another manufacturer were acquired. Additionally, Disposition Services would have inconsistent evidential matter for audit responses, which would be nonresponsive to audit, if another manufacturer's XRF instruments and associated equipment were procured.

DLA Strategic Materials can only accept reading/results from Bruker Titan 800 XRF Guns. This is a DLA Strategic Materials requirement for anything DLA Disposition Services is processing for reclamation /deposit into DLA's Strategic and Critical Materials listing.

DLA Strategic Materials Chemists are trained in XRF Gun calibration and Matrix / Spectrum results and analysis. Having the same XRF Gun used across DLA Streamlines processing strategic, critical and rare earth elements for recovery. Additionally, having these XRF guns at our field sites will enable Disposition Services Sales to list the alloy characteristics in the sales descriptions and drive-up sales revenue.

DLA Disposition Services has an established business process with DLA Strategic Materials from end item identification through shipment to designated DLA contractor "if" the Bruker Titan 800 is used due to trusted results and the internal capability to provide chemical result verification when needed.

Having all DLA Disposition Services sites using the same XRF Gun streamlines both the training / certification requirements as well as characteristic results/reading from usage. Items scanned by Disposition Services personnel which meet the criteria will be direct shipped to DLA Contractors (Germanium Recovery, Gallium Recovery and Titanium Processing). Using other manufactures will "not" benefit DLA because the only accepted items are required to be scanned by the Bruker Titan 800, if

another manufacture is used then DLA Strategic Materials will be unable to accept our reading/results and will be required to travel (TDY) to establish a Bruker Titan 800 reading.

Unlike other manufacturers of hand-held XRF instruments, the radiation source for the Bruker instrument is generated electronically. Other manufacturers of these instruments use collimators to focus radiation from a radionuclide contained within the instrument. What this means is, in the event a Bruker instrument is accidentally destroyed, there is zero risk of a radioactive event at the depot or at any other site where Disposition Services personnel would be using the Bruker instrument. If we had a similar instrument from another manufacturer, there would be a radionuclide present within the housing of the instrument. In the event of accidental destruction, release of the radionuclide would result in a radiation event and safety issue for DLA personnel. A radiation event means multiple agencies, including both state and federal, would shut down the depots and shut down Disposition Services operations.

An alloy library of spectra for super alloys is produced by Bruker and built into the Bruker instrument. The Bruker alloy library is the best match to DLA Strategic Materials inventory compared to all other alloy libraries from other XRF instrument manufacturers. Thus, alloy libraries from manufacturers, other than Bruker, of hand-held XRF instruments could not be used with existing DLA scans of alloys, and would create an auditing issue because it is not comparable to the existing data DLA has on its alloy materials. The result would be that DLA would have inconsistent data. Additionally, DLA would have inconsistent evidential matter for audit response, which would be nonresponsive to audit.

The Bruker instrument covers 37 elements, and includes the lighter elements Ca, Al, Si, and Mg. This broad range of element coverage is required by DLA and is not met by other hand-held XRF instrument manufacturers.

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